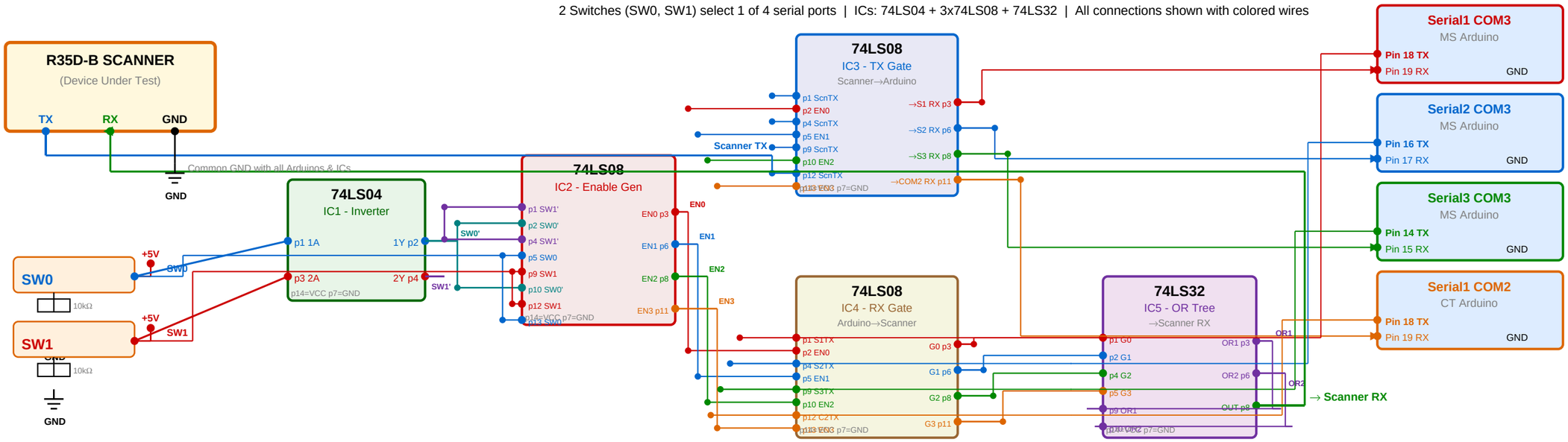
# SERIAL PORT SELECTOR — WIRING DIAGRAM

2 Switches (SW0, SW1) select 1 of 4 serial ports | ICs: 74LS04 + 3x74LS08 + 74LS32 | All connections shown with colored wires




**WIRE COLOR LEGEND:**

- Scanner TX
- Scanner RX (from OR tree)
- SW0 / SW0 direct
- SW1 / SW1 direct
- SW0' (inverted)
- SW1' (inverted)
- EN0 / Serial1 COM3
- EN1 / Serial2 COM3
- EN2 / Serial3 COM3
- EN3 / Serial1 COM2

NOTE: All IC VCC=Pin14 (+5V), GND=Pin7. Add 0.1µF bypass cap on each IC. All GNDs connected together.

# PIN-TO-PIN CONNECTION TABLE

Every physical wire listed: FROM → TO | Use this as your wiring checklist

| # | Wire Color | FROM Component | FROM Pin | TO Component | TO Pin | |
|---|---|---|---|---|---|---|
| 1 | Blue | SW0 Switch | Output terminal | IC1 74LS04 | Pin 1 (1A input) | |
| 2 | Red | SW1 Switch | Output terminal | IC1 74LS04 | Pin 3 (2A input) | |
| 3 | Teal | IC1 74LS04 | Pin 2 (1Y = SW0') | IC2 74LS08 | Pin 2 (1B) + Pin 10 (3B) | |
| 4 | Purple | IC1 74LS04 | Pin 4 (2Y = SW1') | IC2 74LS08 | Pin 1 (1A) + Pin 4 (2A) | |
| 5 | Blue | SW0 Switch | Output terminal (direct) | IC2 74LS08 | Pin 5 (2B) + Pin 13 (4B) | |
| 6 | Red | SW1 Switch | Output terminal (direct) | IC2 74LS08 | Pin 9 (3A) + Pin 12 (4A) | |
| 7 | DkRed | IC2 74LS08 | Pin 3 (EN0 output) | IC3 Pin 2 + IC4 Pin 2 | (2B on both) | |
| 8 | Blue | IC2 74LS08 | Pin 6 (EN1 output) | IC3 Pin 5 + IC4 Pin 5 | (3B on both) | |
| 9 | Green | IC2 74LS08 | Pin 8 (EN2 output) | IC3 Pin 10 + IC4 Pin 10 | (3B on both) | |
| 10 | Orange | IC2 74LS08 | Pin 11 (EN3 output) | IC3 Pin 13 + IC4 Pin 13 | (4B on both) | |
| 11 | Blue | R35D-B Scanner | TX wire (Yellow) | IC3 74LS08 | Pin 1 + 4 + 9 + 12 (all A inputs) | |
| 12 | DkRed | IC3 74LS08 | Pin 3 (1Y output) | MS Arduino COM3 | Pin 19 (Serial1 RX) | |
| 13 | Blue | IC3 74LS08 | Pin 6 (2Y output) | MS Arduino COM3 | Pin 17 (Serial2 RX) | |
| 14 | Green | IC3 74LS08 | Pin 8 (3Y output) | MS Arduino COM3 | Pin 15 (Serial3 RX) | |
| 15 | Orange | IC3 74LS08 | Pin 11 (4Y output) | CT Arduino COM2 | Pin 19 (Serial1 RX) | |
| 16 | DkRed | MS Arduino COM3 | Pin 18 (Serial1 TX) | IC4 74LS08 | Pin 1 (1A input) | |
| 17 | Blue | MS Arduino COM3 | Pin 16 (Serial2 TX) | IC4 74LS08 | Pin 4 (2A input) | |
| 18 | Green | MS Arduino COM3 | Pin 14 (Serial3 TX) | IC4 74LS08 | Pin 9 (3A input) | |
| 19 | Orange | CT Arduino COM2 | Pin 18 (Serial1 TX) | IC4 74LS08 | Pin 12 (4A input) | |
| 20 | DkRed | IC4 74LS08 | Pin 3 (1Y = G0_RX) | IC5 74LS32 | Pin 1 (1A input) | |
| 21 | Blue | IC4 74LS08 | Pin 6 (2Y = G1_RX) | IC5 74LS32 | Pin 2 (1B input) | |
| 22 | Green | IC4 74LS08 | Pin 8 (3Y = G2_RX) | IC5 74LS32 | Pin 4 (2A input) | |
| 23 | Orange | IC4 74LS08 | Pin 11 (4Y = G3_RX) | IC5 74LS32 | Pin 5 (2B input) | |
| 24 | Purple | IC5 74LS32 | Pin 3 (1Y = OR1) | IC5 74LS32 | Pin 9 (3A input) | |
| 25 | Purple | IC5 74LS32 | Pin 6 (2Y = OR2) | IC5 74LS32 | Pin 10 (3B input) | |
| 26 | Green | IC5 74LS32 | Pin 8 (3Y = final output) | R35D-B Scanner | RX wire (Green) | |
| 27 | Red | +5V Power Rail | VCC | IC1-IC5 | Pin 14 on each IC | |
| 28 | Black | GND Rail | Ground | IC1-IC5 | Pin 7 on each IC | |
| 29 | Black | GND Rail | Ground | R35D-B Scanner | GND wire (Black) | |
| 30 | Black | GND Rail | Ground | Both Arduinos | GND pin | |
| 31 | Red | +5V Power Rail | VCC | SW0 Switch | One terminal (other to signal+10kΩ pulldown) | |
| 32 | Red | +5V Power Rail | VCC | SW1 Switch | One terminal (other to signal+10kΩ pulldown) | |

## SWITCH SELECTION TABLE:

| SW1 | SW0 | EN Active | Selected Port | Arduino | TX Pin | RX Pin |
|---|---|---|---|---|---|---|
| 0 (OFF) | 0 (OFF) | EN0 | Serial1 COM3 | MS Arduino | Pin 18 | Pin 19 |
| 0 (OFF) | 1 (ON) | EN1 | Serial2 COM3 | MS Arduino | Pin 16 | Pin 17 |
| 1 (ON) | 0 (OFF) | EN2 | Serial3 COM3 | MS Arduino | Pin 14 | Pin 15 |
| 1 (ON) | 1 (ON) | EN3 | Serial1 COM2 | CT Arduino | Pin 18 | Pin 19 |